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Combining mental and physical practice using brain computer interface driven hand exoskeleton for personalized post-stroke neuro-rehabilitation

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1. Introduction

Every year a massive number of people get affected by stroke with an annual estimate of over 15 million and up to 40% of stroke survivors may suffer from permanent upper limb paralysis, which may significantly impact their quality of life and employability. Often the upper limb paralysis becomes chronic due to lack of active and engaging rehabilitation exercises over a prolonged period [1]. This research introduces a novel neuro-rehabilitation system consisting of an advanced Brain-Computer Interface (BCI) for providing mental practice (MP) in conjunction with an exoskeleton based physical practice (PP) for neuro-rehabilitation of stroke patients.

2. Material and Methods

The proposed neuro-rehabilitation system facilitates both MP and PP using an EEG based BCI [2, 3] and hand exoskeleton. The exoskeleton provides active PP and also proprioceptive neurofeedback. The feasibility of the system has been tested by conducting a 6 week long clinical trial on 4 hemiplegic stroke patients suffering from left hand disability. In total 12 to 16 sessions were held, each session comprised of 30 min of hand-exoskeleton assisted PP followed by 30 min of MP. During MP patients were provided with contingent visual feedback using a gradual hand extension video combined with proprioceptive feedback of their impaired hand extension using the hand exoskeleton. The rehabilitation outcomes were measured using grip strength (GS) and Action Research Arm Test (ARAT). The experimental environment and the exoskeleton used are shown in Fig.1 and Fig. 2, respectively. The proposed experimental paradigm of BCI system is depicted in Fig. 3.

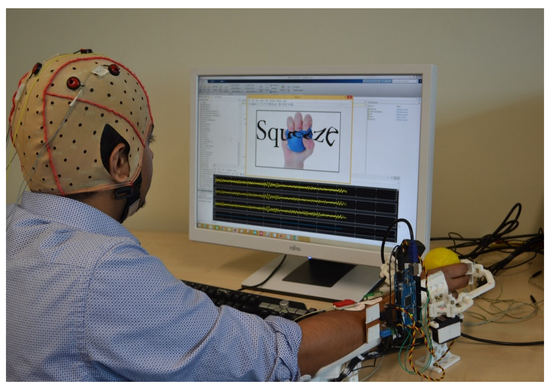

Fig. 1. Experimental environment: Participant controlling hand exoskeleton using the proposed BCI paradigm.

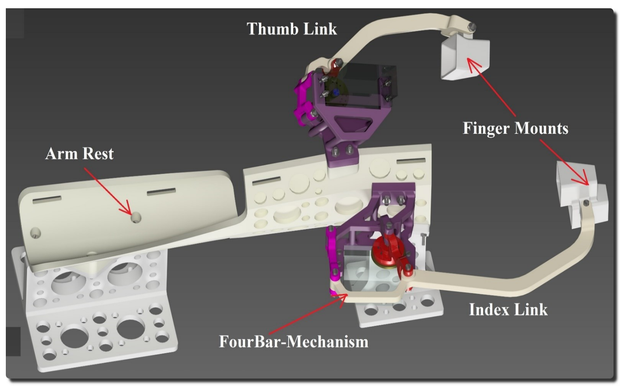

Fig. 2. The CAD model of the developed hand exoskeleton. The arm rest is attached for the convenience of the user; finger mounts at the end of thumb and index link are for the placement of user's fingertip.

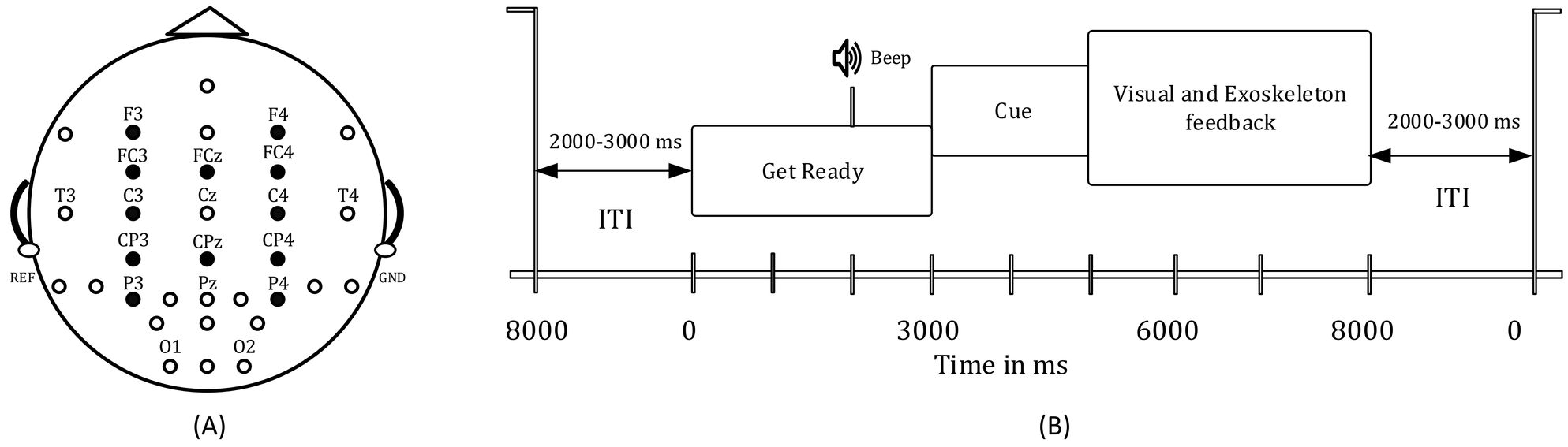

Fig. 3. The proposed experimental paradigm of BCI system. (A) Electrode montage according to International 10-20 system, the channels F3, F4, FC3, FCz, FC4, C3, C4, CP3, CPz, CP4, P3, and P4 are selected, which are represented by black circles. (B) The timing diagram of the experimental paradigm during neurofeedback stage.

3. Results

Three out of four patients demonstrated significant improvement in both GS and ARAT scores when compared to the baseline (see Fig.4 and Fig.5), while two of them achieved minimal clinically important difference (MCID) in their ARAT scores. The classification accuracy (CA) of the BCI also increased through therapy sessions and it also highly correlated with the outcome measures (see Fig. 6).

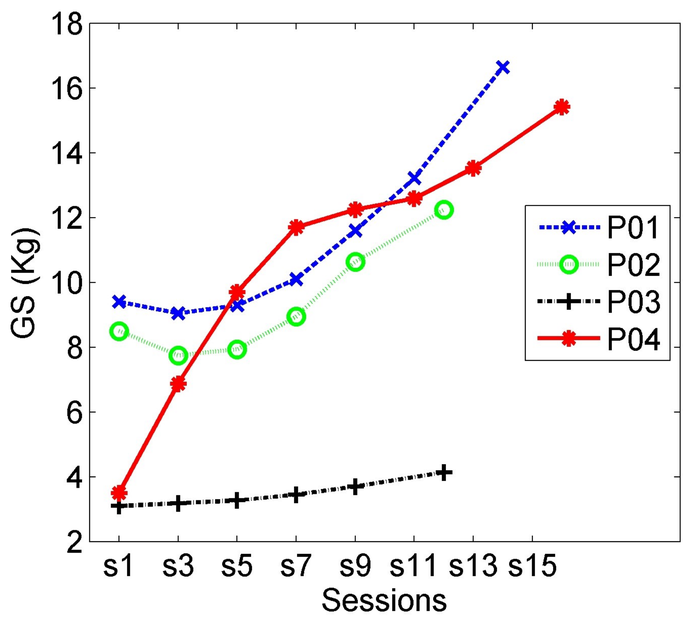

Fig. 4. Change in the grip strength (GS) through the therapy sessions.

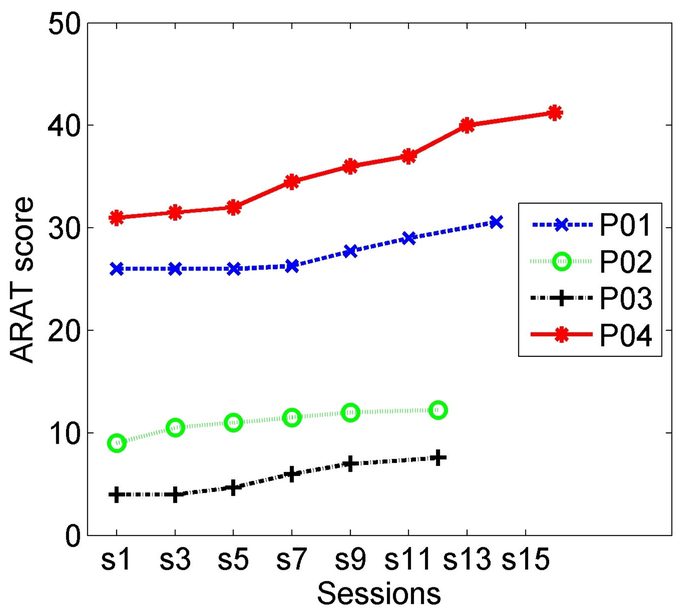

Fig. 5. Change in the Action Research Arm Test (ARAT) through the therapy sessions.

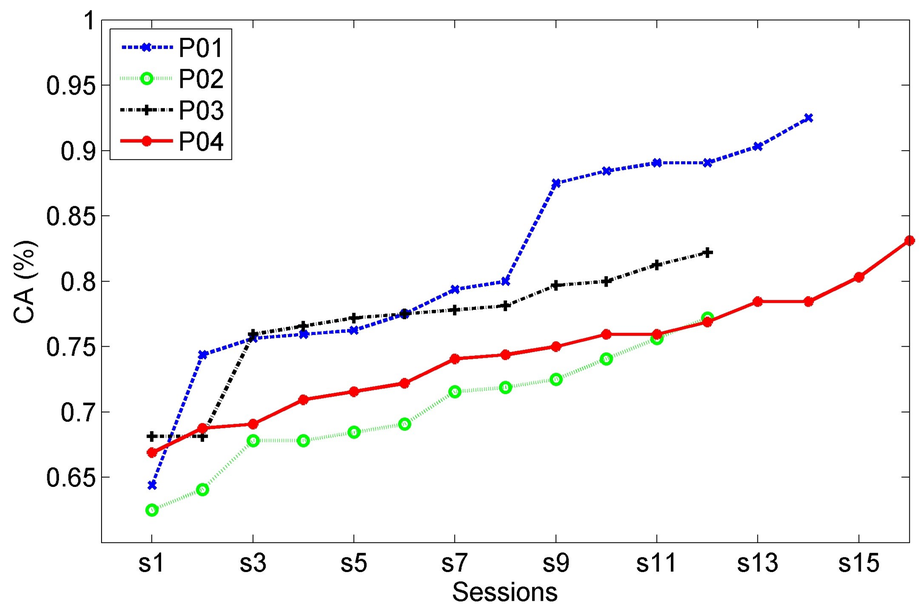

Fig. 6. Change in the classification accuracy (CA) through the therapy sessions.

4. Conclusion

Considering the positive outcomes of the study, it can be concluded that the proposed rehabilitation system is an effective method for devising a combined MP and PP based post-stroke rehabilitation strategy for hand functional recovery. There were high rates of attendance and no negative effects in terms of changes in mood or fatigue and the feedback from users was positive. Although these findings are promising and render the proposed neuro- rehabilitation system feasible, the scope of the final conclusions is limited by the small sample size and the lack of a control group.

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Acknowledgment

This research work is supported by the UKIERI DST Thematic Partnership project "A BCI operated hand exoskeleton based neuro-rehabilitation system" (UKIERI-DST-2013-14/126).

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